



USER MANUAL
NASCAR GEN 4 CARS



TABLE OF CONTENTS

CLICK TO VIEW A SECTION

GENERAL INFORMATION	
<i>A Message From iRacing »</i>	3
<i>Tech Specs »</i>	4
<i>Introduction »</i>	5
Getting Started »	5
Loading An iRacing Setup »	6
<i>Dash Pages »</i>	7
Page 1 »	7
Page 2 »	8
ADVANCED SETUP OPTIONS	
<i>Tires »</i>	10
Tire Data »	10
<i>Chassis »</i>	11
Front »	11
Front ARB »	12
Front Corners »	13
Rear Corners »	15
Rear »	17





DEAR iRACING USER,

Congratulations on your purchase of a NASCAR 2003 Gen 4 car! From all of us at iRacing, we appreciate your support and your commitment to our product. We aim to deliver the ultimate sim racing experience, and we hope that you'll find plenty of excitement with us behind the wheel of your new car!

The following guide explains how to get the most out of your new car, from how to adjust its settings off of the track to what you'll see inside of the cockpit while driving. We hope that you'll find it useful in getting up to speed.

Thanks again for your continued support, and we'll see you on the track!



CHASSIS

DOUBLE WISHBONE INDEPENDENT
FRONT / LIVE AXLE TRUCK ARM REAR



LENGTH
5156 mm
203 in

WIDTH
1905 mm
75 in

WHEELBASE
2794 mm
110 in

DRY WEIGHT
1542 kg
3400 lbs

WET WEIGHT
WITH DRIVER
1657 kg
3654 lbs

POWERED UNIT

NATURALLY ASPIRATED
STEEL BLOCK PUSHROD V8



DISPLACEMENT
5.9 Liters
358 cid

STANDARD

RPM LIMIT
9800

TORQUE
550 lb-ft
740 Nm

POWER
850 bhp
633 kW

RESTRICTED

RPM LIMIT
8000

TORQUE
390 lb-ft
530 Nm

POWER
450 bhp
335 kW



INTRODUCTION

The information found in this guide is intended to provide a deeper understanding of the chassis setup adjustments available in the garage, so that you may use the garage to tune the chassis setup to your preference.

Before diving into chassis adjustments, though, it is best to become familiar with the car and track. To that end, we have provided baseline setups for each track commonly raced by these cars.

To access the baseline setups, simply open the Garage, click iRacing Setups, and select the appropriate setup for your track of choice. If you are driving a track for which a dedicated baseline setup is not included, you may select a setup for a similar track to use as your baseline.

After you have selected an appropriate setup, get on track and focus on making smooth and consistent laps, identifying the proper racing line and experiencing tire wear and handling trends over a number of laps.

Once you are confident that you are nearing your driving potential with the included baseline setups, read on to begin tuning the car to your handling preferences.

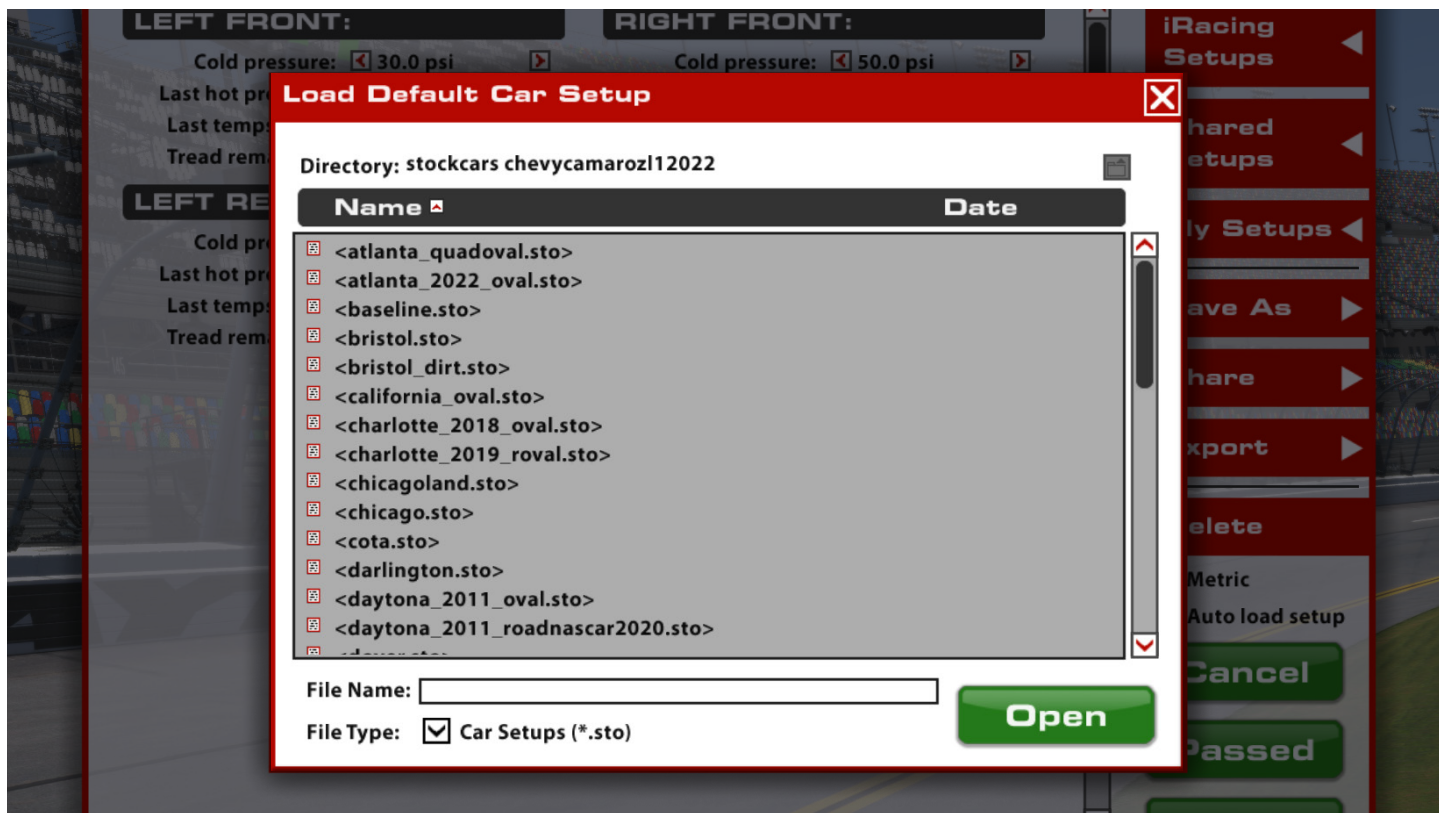
GETTING STARTED



Once you load into the car, select 1st gear. Slowly release the clutch while applying the throttle to drive away. A clutch is necessary when coming to a stop to prevent stalling the engine and shifting into reverse if necessary, but the clutch isn't required once the vehicle is in motion for upshifts or downshifts.

To upshift, simply let off the throttle and select the next higher gear. To downshift, give the throttle a blip while selecting the next lower gear. If you downshift too early, or don't blip the throttle sufficiently, the wheel speed and engine speed will be mismatched, leading to wheel hop at the rear and a possible spin.

LOADING AN iRACING SETUP



When you first load into a session, the iRacing Baseline setup will be automatically loaded onto the car. If you would like to try any of the other iRacing pre-built options, you may select it by going to Garage > iRacing Setups > and then selecting another option that fits your needs.

Because this car uses slightly different chassis and body configurations on different types of tracks, it will be necessary to load a setup from the same track type to pass tech inspection. For example, a setup for Talladega will pass at Daytona, but likely will not pass at Bristol.

If you would like to customize the setup, simply make the changes in the garage that you would like to update and click apply. If you would like to save your setup for future use click "Save As" on the right to name and save the changes.

To access all of your personally saved setups, click "My Setups" on the right side of the garage. If you would like to share a setup with another driver or everyone in a session, you can select "Share" on the right side of the garage to do so. If a driver is trying to share a setup with you, you will find it under "Shared Setups" on the right side of the garage as well.

To save your setup for reference later, click the "Export" button on the right side of the garage and save the file. This will produce an HTML file with all of the settings for that setup that can be opened at any time and viewed in a web browser.

DASH PAGES

The 2003 Gen 4 cars feature full sets of analog gauges typical for that era. The Chevrolet and the Ford have unique layouts but display the same information.

CHEVROLET



The Monte Carlo places the four system gauges on the left in a cluster, the tachometer to the right of the steering column, and a voltmeter to the right.

Oil Pressure - Engine Oil Pressure, in psi

Fuel Pressure - Fuel system pressure, in psi

Oil Temperature - Engine Oil Temperature, in °F

Water Temperature - Engine cooling water temperature, in °F

Tachometer - Engine RPM

Voltmeter - Electrical system voltage

FORD



From left to right:

Tachometer - Engine RPM

Fuel Pressure - Fuel system pressure, in psi

Water Temperature - Engine cooling water temperature, in °F

Oil Pressure - Engine Oil pressure, in psi

Oil Temperature - Engine Oil temperature, in °F

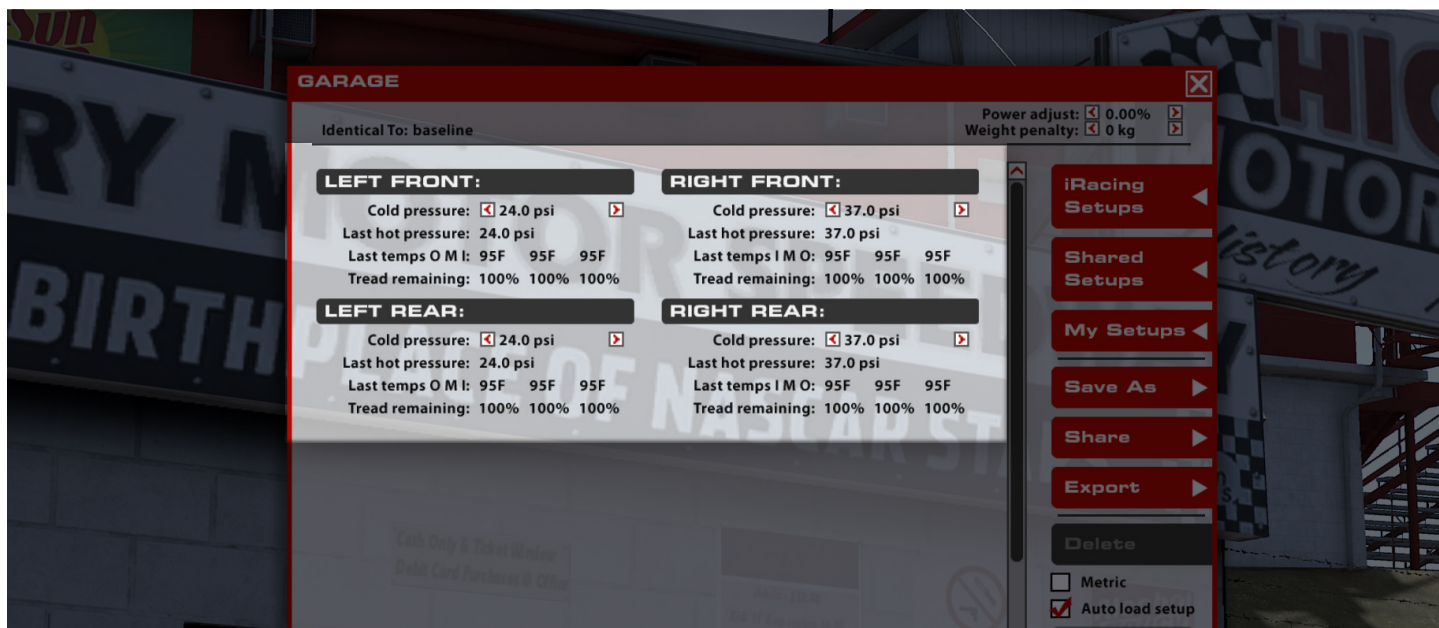
Voltmeter - Electrical system voltage

ADVANCED SETUP OPTIONS

This section is aimed toward more advanced users who want to dive deeper into the different aspects of the vehicle's setup. Making adjustments to the following parameters is not required and can lead to significant changes in the way a vehicle handles. It is recommended that any adjustments are made in an incremental fashion and only singular variables are adjusted before testing changes.

TIRES

TIRE DATA



COLD PRESSURE / STARTING PRESSURE

The air pressure in the tires when the car is loaded into the world. Lower pressures will provide more grip but will produce more rolling drag and build temperature faster. Higher pressures will feel slightly more responsive and produce less rolling drag, but will result in less grip. Generally, higher pressures are preferred at tracks where speeds and loads are higher while lower pressures work better at slower or flatter tracks where mechanical grip is important.

LAST HOT AIR PRESSURE

When the car returns to the garage after running laps on the track, the measured tire pressure will be displayed as Hot Pressure. The difference between cold and hot pressure is a good way to see how tires are being loaded and worked while on track. Tires seeing more work and load will build more pressure, and paying attention to which tires are building more pressure and adjusting cold pressure to compensate can be crucial for optimizing tire performance. Hot Pressures will usually plateau into a run, these peak pressures heavily influencing the car's balance through the run. By adjusting cold pressures based on differences in the Hot Pressures, long-run balance can be tuned to prevent changes in handling.

LAST TEMPERATURES

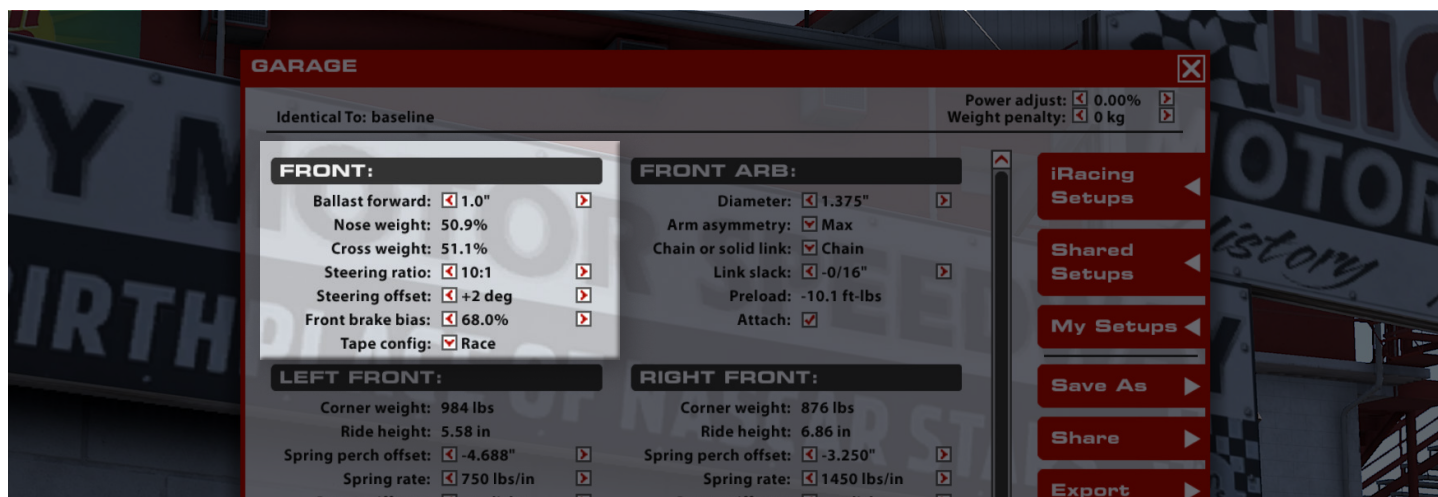
The tire carcass temperatures (measured within the tread) are displayed after the car returns from the track. These temperatures are an effective way to determine how much work or load a given tire is experiencing while on track. Differences between the inner and outer temperatures can be used to tune individual wheel alignment and the center temperatures can be compared to the outer temperatures to help tune tire pressure.

TREAD REMAINING

The amount of tread on the tire, displayed as a percentage of a new tire, is shown below the tire temperatures. These values are good for determining how far a set of tires can go before needing to be replaced, but don't necessarily indicate an under- or over-worked tire in the same way temperatures will.

CHASSIS

FRONT



BALLAST FORWARD

To meet minimum weight requirements, tungsten blocks are installed within the lower frame rails on the chassis. These blocks can be moved fore and aft in the chassis, directly influencing the car's Nose Weight value. The Ballast Forward value is simply a measurement of the location of these tungsten blocks relative to a reference point in the frame rail. Moving ballast forward in the car raises Nose Weight, moving it rearward reduces Nose Weight.

NOSE WEIGHT

The vehicle's Nose Weight is the percentage of total vehicle weight on the front tires. Nose Weight represents a rough approximation of the longitudinal Center of Gravity location in the vehicle and has a direct influence on the high-speed stability of the vehicle. Higher Nose Weight values result in a more directionally-stable vehicle, good for tracks with low grip or very high speeds. Conversely, lower Nose Weight values are good for tracks with high grip levels or lower speeds.

CROSS WEIGHT

Cross Weight is the amount of weight on the car's Left-Rear and Right-Front tires relative to the entire weight of the car, displayed in percent. This is adjusted via the corner Spring Perch Offset adjustments as well as ARB preload to a small extent. For an oval car, Cross Weight is one of the most influential settings for grip level while the vehicle is in a turn. Higher Cross Weight values will add weight to the left-rear and right-front, both stabilizing entry and helping drive-off on corner exit. Lower Cross-Weight values will help the vehicle rotate and keep it "free" in the corner to prevent speed from being lost, however too low can result in unstable entry and exit.

STEERING RATIO

The Steering Ratio is a numerical value for how fast the steering response is to driver inputs at the steering wheel. This ratio can be thought of as the degrees of steering input needed to produce one degree of turn on the steering box output shaft. For example, a 12:1 steering ratio will require 12° of steering input to rotate the steering output shaft 1°. A steering box with a lower ratio will feel more responsive to steering inputs and will require less steering input to reach the amount needed to navigate a corner. A steering box with a higher ratio will feel less responsive and will require more steering input to reach the amount needed to navigate a corner.

STEERING OFFSET

Degrees of steering wheel offset, achieved with a combination of installing the steering wheel into the quick release mechanism off-center and adjusting front tie-rods. This can be used to compensate for chassis settings which place the wheel off center and is primarily a driver comfort adjustment.

FRONT BRAKE BIAS

Brake Bias is the percentage of braking force that is being sent to the front brakes. Values above 50% result in more pressure being sent to the front, while values less than 50% send more force to the rear. This should be tuned for both driver preference and track conditions to get the optimum braking performance for a given situation.



TAPE CONFIG

Grille Tape is applied to the radiator openings in the front fascia to reduce drag when the full opening isn't necessary to keep the engine at optimum running temperatures. The Race configuration will open up the radiator opening to allow air into the radiator for race

sessions or long runs in practice. For Qualifying, where performance is key, the Qual setting will fully block the radiator opening. This will dramatically reduce drag and increase downforce, but will cause the engine to overheat very quickly and lead to engine failure. The Qual setting is not available in Race sessions.

FRONT ARB



DIAMETER

The ARB (Anti-Roll Bar) diameter influences the stiffness of the front suspension in roll. Increasing the ARB size will increase the roll stiffness of the front suspension, resulting in less body roll and improved aerodynamic stability but increasing mechanical understeer. This can also, in some cases, lead to a more responsive steering feel for the driver. Conversely, reducing the ARB size will soften the suspension in roll, increasing body roll and reducing aerodynamic efficiency but decreasing mechanical understeer. This can result in a less-responsive feel from the steering.

ARM ASYMMETRY

The difference in length between the left and right sway bar arms can be altered via the Arm Asymmetry settings. The "None" setting will set the two arms at equal length, while increasing the setting will increase the difference in length of the two arms. This can be used to produce multiple effects, primarily serving to produce a higher anti-roll force on the right-front suspension than on the left-front, effectively rolling the chassis to the left when under load. This can be used to correct excessive roll without increasing the ARB diameter. A knock-on effect of asymmetry is a slight increase in front end heave stiffness, or resistance to vertical travel. Since the two different lengths of arms cause the bar to be twisted at different rates, vertical travel will load the ARB, possibly leading to higher front ride heights on straights.

CHAIN / SOLID LINK

The type of linkage installed on the right-side ARB arm can be changed to improve the chassis response to bumps in the track surface. If a Solid link is installed, the ARB will transfer loads from one wheel to the other in all situations. If a Chain is installed, a sharp

bump to the left-front wheel will collapse the right-side linkage and prevent the sudden load from being applied to the right-front wheel. This is very useful over bumpy surfaces where a bump could unload the front tires and cause sudden understeer. Under normal conditions the two settings are identical and will have no influence on the suspension's roll stiffness.

LINK SLACK

The left-side sway bar linkage can be adjusted to either delay bar engagement or apply a static load to the bar. Positive or negative values are referenced as an offset to a standard length and don't necessarily represent anything specific. This adjustment directly affects the bar's Preload, outlined below.

PRELOAD

The ARB Preload is the static load in the bar while the vehicle is in the garage. Preload adjustments can be used to alter the dynamic loads in the bar while on track, and can be used to remove or add bar load in the corners and on the straights. Negative preload will load the bar, causing the chassis to roll to the left under static conditions. If the preload is zero with a positive link slack, the suspension must travel before engaging the ARB. This setting does not affect roll stiffness, however it does affect total bar load while on track.

ATTACH

A quick way to unhook the anti-roll bar to allow for static suspension adjustments without bar twist confusing things; increase link slack and unhook the ARB before making spring/ride height adjustments; attach and reduce link slack (ARB preload) when done. This option will completely detach the bar and keep it from being loaded while on track, if desired.



FRONT CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the Spring Perch Offset setting.

RIDE HEIGHT

Distance from ground to a reference point on the chassis. Front heights are measured at the bottom of the chassis frame rail just behind the wheel well and can be roughly identified via the skirt rivets at the bottom of the door. Since these values are measured to a specific reference point on the car, these values may not necessarily reflect the vehicle's ground clearance, but instead provide a reliable value for the height of the car off of the race track at static values. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's aerodynamic performance as well as mechanical grip.

SPRING PERCH OFFSET

Spring perch offset is used to adjust ride height and corner weight. Adjusting this setting changes the preload on the spring under static conditions. Decreasing the value increases preload on the spring, adding weight to its corner and increasing the ride height at that corner. Increasing the value does the opposite, reducing height and weight on a given corner. These should be adjusted in pairs (left and right, for example) or with all four spring preload adjustments in the car to prevent crossweight changes while adjusting ride height.

SPRING RATE

Spring Rate changes how stiff the spring is, represented in a force per unit of displacement. Primarily responsible for maintaining ride height and a good attitude under changing wheel loads, stiffer springs will maintain the car's attitude better while sacrificing

mechanical grip. Softer springs will deal with bumps better and increase mechanical grip, however this could cause the chassis to pitch and roll too much, hurting aerodynamic performance at faster tracks.

BUMP STIFFNESS

Bump stiffness affects how resistant the shock is to compression (reduction in length), usually in chassis movements as a result of driver input (steering, braking, & throttle) and cornering forces. Higher values will increase compression resistance and transfer load onto a given tire more quickly. Lower values reduce compression resistance and transfer load onto a given tire more slowly. Differences between left and right bump stiffness influence dynamic crossweight on corner entry and result in a dynamic shift in balance while the shocks are being compressed, with greater right front bump shifting the balance toward understeer.

REBOUND STIFFNESS

Rebound stiffness affects how resistant the shock is to extension (increase in length), typically during body movement as a result of driver inputs. Higher rebound values will slow extension of the shock, lower values will allow the shock to extend faster. Higher rebound values can better control aerodynamic attitude but can result in the wheel being unloaded when the suspension can't extend enough to maintain proper contact with the track. When tuning for handling, higher front rebound can increase on-throttle mechanical understeer (but reduce splitter lift, maintaining downforce) while lower values will maintain front end grip longer, helping to reduce mechanical understeer, but will allow more splitter lift and reduce downforce. Differences between left and right front rebound influence dynamic cross on corner exit, with greater right front rebound shifting the balance toward oversteer. Excessive front rebound can lead to unwanted oscillations due to the wheel bouncing off of the track surface instead of staying in contact.



CAMBER

Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber is when the top of the wheel is closer to the chassis centerline than the bottom of the wheel, positive camber is when the top of the tire is farther out than the bottom. Higher camber angles will increase the cornering force generated by the tire, but will reduce the amount of longitudinal grip the tire will have under braking. Excessive camber values can produce very high cornering forces but will also significantly reduce tire life, so it is important to find a balance between life and performance. For ovals, set the left side positive and the right side negative. For road courses, all four wheels should be set with negative camber.

CASTER

Caster is how much the steering axis is leaned back (positive) or forward (negative), which influences dynamic load jacking effects as the car is steered. More positive caster results in a heavier steering feel but decreases dynamic crossweight while turning, as well as adding straight-line stability. Running less caster on the left-front will cause the vehicle to pull to the left, a desirable effect on ovals.

TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Positive toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and negative toe-in (toe-out) is when the front of the wheel is farther away from the centerline than the rear of the wheel. More net toe-out will destabilize the car in a straight line and help with turn-in but increases slip angle on the front wheels and can risk over-slipping the front tires at high values, which can result in sudden understeer when cornering. On ovals, front toe will usually be different for each wheel due to asymmetric loading on the front tires.



REAR CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the Spring Perch Offset setting.

RIDE HEIGHT

Distance from ground to a reference point on the chassis. Since these values are measured to a specific reference point on the car, these values may not necessarily reflect the vehicle's ground clearance, but instead provide a reliable value for the height of the car off of the race track at static values. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's aerodynamic performance as well as mechanical grip. Increasing rear ride height will shift aero forward (more oversteer) as well as slightly increasing overall downforce and drag. Conversely, reducing rear ride height will increase rear and overall downforce and reduce drag.

SPRING PERCH OFFSET

Used to adjust ride height and corner weight, adjusting this setting applies a preload to the spring under static conditions. Decreasing the value increases preload on the spring, adding weight to its corner and increasing the ride height at that corner. Increasing the value does the opposite, reducing height and weight on a given corner. These should be adjusted in pairs (left and right, for example) or with all four spring preload adjustments in the car to prevent crossweight changes while adjusting ride height.

SPRING RATE

Spring Rate changes how stiff the spring is, represented in force per unit of displacement. Primarily responsible for maintaining ride height and aerodynamic attitude under changing wheel loads, stiffer springs control the chassis attitude better (less roll or pitch change) which is good for aerodynamics and camber control, but mechanical grip is often better with softer springs which allow for more track surface compliance but reduce aerodynamic control. For ovals, a softer left-rear spring (relative to the right-rear) is desired to prevent the dynamic cross from being too high in the corners, which will result in a balance shift towards understeer through a run.

BUMP STIFFNESS

Bump stiffness affects how resistant the shock is to compression (reduction in length), usually in chassis movements as a result of driver input (steering, braking, & throttle) and cornering forces. Higher values will increase compression resistance and transfer load onto a given tire more quickly. Lower values reduce compression resistance and transfer load onto a given tire more slowly. Differences between left and right bump stiffness influence dynamic crossweight and result in a dynamic shift in balance while the shocks are being compressed, particularly on corner exit, with greater right rear bump shifting the dynamic balance toward oversteer.



REBOUND STIFFNESS

Rebound stiffness affects how resistant the shock is to extension (increase in length), typically during body movement as a result of driver inputs. Higher rebound values will slow extension of the shock, lower values will allow the shock to extend faster. Higher rebound values can better control aerodynamic attitude but can result in the wheel being unloaded when the suspension can't extend enough to maintain proper contact with the track. When tuning for handling, higher rear rebound can increase off throttle mechanical oversteer while lower values will maintain rear end grip longer, helping to reduce mechanical oversteer off throttle, but will allow more rear deck lift and shift the aero balance forward. Differences between left and right rear rebound can influence off throttle dynamic cross weight on corner entry, with greater right rear rebound shifting the dynamic balance toward understeer.

TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Positive toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and negative toe-in (toe-out) is when the front of the wheel is farther away from the centerline than the rear of the wheel. At the rear, toe-out will help with turn in and direction changes, while toe-in will increase straight-line stability. For short tracks or tracks where traction is a problem, setting both rear wheels toed to the left (positive right-rear and negative left-rear) can help to straighten the car on throttle application.

CAMBER

Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber is when the top of the wheel is closer to the chassis centerline than the bottom of the wheel, positive camber is when the top of the tire is farther out than the bottom. Greater camber angles will increase the cornering force generated by the tire, but will reduce the amount of longitudinal grip the tire will have under braking. Excessive camber values can produce very high cornering forces but will also significantly reduce tire life, so it is important to find a balance between life and performance. For ovals, set the left side positive and the right side negative. For road courses, both rear wheels should be set with negative camber.

TRACK BAR HEIGHT

The Track Bar is a solid steel rod mounted to the rear axle housing on the left side and to the chassis frame on the right side to hold the rear axle assembly in place. The track bar's left- and right-side mounts can be raised or lowered to alter both the dynamic loading on the rear tires as well as change the rear suspension's roll stiffness, with both effects having a large effect on handling characteristics.

Raising the overall height of the track bar (equal adjustment on both ends of the bar in the same direction, e.g. 1" up on both ends) will increase the roll stiffness of the rear suspension and free up the chassis through all phases of cornering. Lowering the track bar on both ends will reduce the rear roll stiffness and tighten the chassis while cornering. Since this type of adjustment alters rear roll stiffness it can have a large effect on the chassis's roll attitude, so it's important to keep this in mind while making adjustments.

Adjusting the two ends individually will shift loads from one rear tire to the other as the chassis loads. Raising the right-side track bar mount will reduce right-rear grip, freeing up the chassis during the transition from center to corner exit, and is a good adjustment for a "tight-center, loose-off" condition. Raising the left-side track bar mount will help the car rotate into the center of the corner. Conversely, lowering one of the mounts will do the opposite and increase grip on its associated tire.

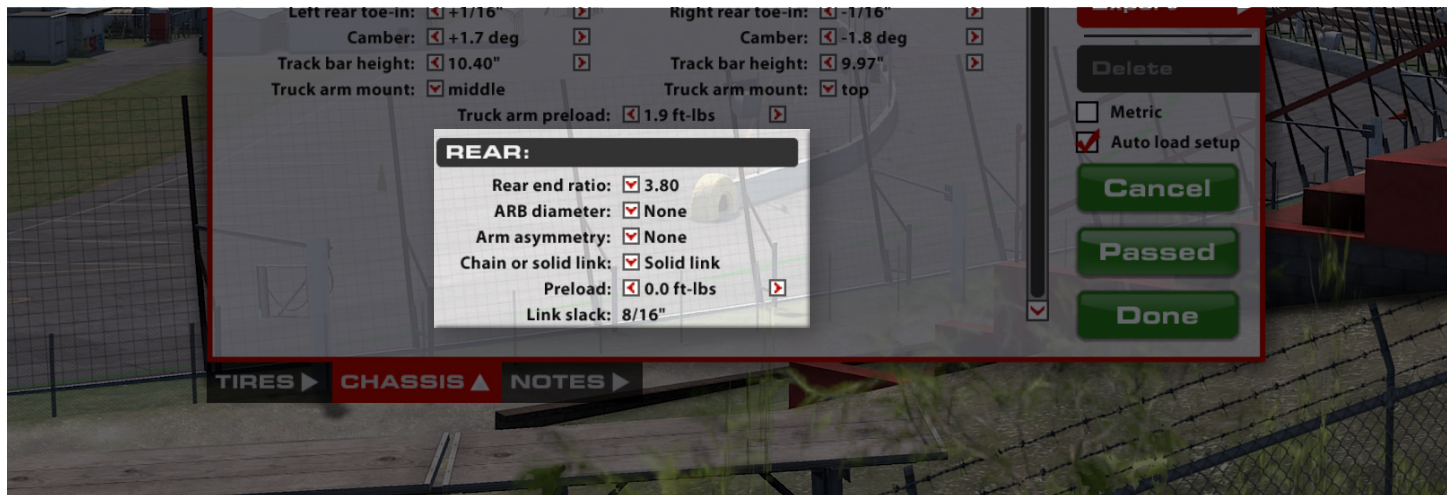
Since the Track Bar Height values are calculated based on Ride Heights, these will change in the garage as other parts of the car are adjusted, however the right-rear mount will move more than the left-rear since that mount is on the chassis itself. Before making adjustments, note the track bar heights and be sure to return them to the starting values after adjustments are completed.

TRUCK ARM MOUNT

The rear axle is held in place longitudinally with two truck arms, mounted to the bottom of the chassis underneath the driver compartment. The forward mounts can be adjusted up and down, resulting in various anti-squat and rear-steer configurations. Higher truck arm mounts will reduce rear end grip, increase rear steer, add anti-squat, and reduce wheel hop under heavy braking. Lower truck arm mounts will increase rear end bite, decrease rear steer, reduce anti-squat, and increase the chances of wheel hop under heavy braking.



REAR



REAR END RATIO

The Rear End Ratio is the ratio between the driveshaft pinion and the differential ring gear. Higher number values produce better acceleration but reduce top speed, lower number values reduce acceleration but result in a higher top speed.

ARB DIAMETER

The Rear Anti-Roll Bar (ARB, or Sway Bar) diameter affects the roll stiffness of the rear suspension. Larger bar diameters will increase rear roll stiffness and can increase steering responsiveness, but can induce oversteer when cornering. Smaller bar diameters will increase body roll as well as reduce oversteer when cornering.

ARM SYMMETRY

As with the front ARB, the lengths of the rear ARB arms can be split to create asymmetric loading across the rear axle. Higher asymmetry values will lengthen the left-side arm, effectively weakening the bar on the left-rear corner compared to the right-rear. This can be used to create a behavior similar to having a stiff right-rear spring in the corners without having it on the straights, very helpful for tracks with high speeds and corner loads but long, fast straights.

CHAIN / SOLID LINK

Similar to the front ARB linkage, the rear linkage can be set to a chain or a solid link. If the chain link is installed, bumps to the left-rear wheel won't always be transferred directly to the right-rear wheel, helping to maintain grip over bumpy track surfaces.

PRELOAD

Preload is the static load in the bar while the vehicle is in the garage. Rear bar preload is useful for tuning its behavior in transitions, like when corner banking is building up or falling away. Higher preload values will keep the bar loaded more through these situations, placing more load onto the right-rear corner and inducing oversteer, but does not affect rear roll stiffness. As with the front ARB, zero preload and a positive link slack will require the suspension to travel before the bar is engaged.

LINK SLACK

The left-side sway bar linkage can be adjusted to either delay bar engagement or apply a static load to the bar. Positive or negative values are referenced as an offset to a standard length and don't necessarily represent anything specific. This adjustment directly affects the bar's Preload, outlined above.

SETUP TIPS

This section is aimed toward helping users who want to dive deeper into the different aspects of the vehicle's setup.



SETUP TIPS

The NASCAR 2003 Gen 4 cars harken back to a simpler era in Stock Car racing, before Coil-Binding, bump stops, and many of the aerodynamic exploits we see in modern times. As a result, the cars are much simpler to understand and set up, providing a great way

to learn the basics. For iRacing Official series, setups are fixed for all cars, but should you wish to run the car in an open-setup League or Hosted event, here are tips on how to get the most out of these high-powered machines.

DRIVING TIPS

The 2003 Gen 4 cars were, in their day, the fastest stock cars on track every weekend. Their high power and low aero create a dramatically different driving behavior than modern cars, and have to be approached that may not be familiar if you're used to modern stock cars.

The most immediate difference is the engine. Since engine power and torque changes with altitude and air temperature, engine performance will change based on the track's location. Tracks at higher altitudes, or races run in hotter air temperatures, will have less power (usually around 850hp) than tracks closer to sea-level, which can surpass 900hp in the right conditions. This is a significant amount of horsepower for the rear tires to handle, so throttle control is crucial for running good lap times. Drivers may notice that, even on a well-balanced chassis, the rear tires can still slip and spin when leaving a corner, even at high speeds. This can be disastrous for drivers who pick up the throttle too soon, induce understeer in the second half of a corner, leading to a spin at corner exit (the typical push-loose condition).

While the aerodynamic performance of these cars was considered quite good in 2003, the overall downforce is relatively low by today's standards. When combined with lower drag and high power, this creates a unique combination of high corner-entry speeds without the available grip to carry speed through the corner. A track like Michigan, for example, can easily exceed 200mph into Turn 1, but the lack of downforce means corner speeds can drop down into the 150-160mph range. This means the driver has to scrub away a lot of speed in a short time, so drivers will need to use a lot of brake and let the car coast through the corner until throttle can be applied. Not enough braking will cause the car to understeer through the center, not enough coasting will lead to a push-loose off the corner.

For drivers new to these cars, it's highly recommended to stick to the provided setups until comfortable with the car's quirks and behavior. Adjusting the car instead of learning to control the throttle can lead to bad habits down the road.

AERODYNAMICS

While aerodynamics was still in relatively early days in 2003, these cars do generate a good bit more downforce than their predecessors without being as sensitive to pitch and height as cars in recent years. Thus, target heights are more vague and the car has a much larger "sweet spot" than newer cars.

On the front end, we've modeled the way aerodynamics were understood in this era. The valence's lower edge, marked as **CFSRRideHeight** in telemetry, should be run around 1 inch from the ground for peak downforce. Raising the front end will reduce overall downforce and shift aero forward slightly, lowering the front end from this point will stall the valence and reduce overall downforce.

At the rear, targets become slightly more complicated thanks to the offset heights of the chassis frame rails. Measuring to the telemetry ride height channels (LR & RRRideHeight), having the left-rear around 4" and the right-rear around 5.5" is a good baseline for most tracks. Generally, running the car at a level attitude, if not slightly rear-up, will be a good configuration. As mentioned, this car is not very pitch sensitive and has a large sweet spot, so it's not as crucial to have the heights perfect when compared to modern cars.

Superspeedways bring a unique challenge with the different body style used at these tracks. Aerodynamic height targets change entirely with this body style, since downforce isn't really necessary in most cases. Where the downforce body wants either a level or rear-up attitude, the superspeedway body operates best when the rear is run lower than the front. Front valence heights (CFSRRideHeight) will work best between 1 and 2 inches above the ground, with higher heights producing slightly less drag and downforce. Lower than this will produce more drag, so make sure the nose stays up! For the rear, heights around 3.5" on the left-rear and 4" on the right-rear should be the targets.



SPRINGS

These cars ran during the transition years between the older Conventional-style setups and newer Big-Bar/Soft-Spring setups, but preceded Coil-Binding as it eventually became known by a couple of years. Spring packages for the 2003 cars will be heavily

track-dependent, with some tracks working better on Conventional-style spring packages (higher-banked, or larger tracks) and others working better with softer front spring packages (shorter and/or flatter tracks).

WEIGHT DISTRIBUTION

The relatively low aerodynamic loads and high amount of suspension travel lend the cars to being very mechanically-dependent, and thus very sensitive to weight placement. Noseweight can be useful in stabilizing the car at high speed tracks like Michigan or Indianapolis but can lead to understeer, while lower noseweight values will help rotate the car around slow, tight corners like Martinsville or Richmond. Superspeedways, where the car is not grip-limited and any understeer will scrub speed away, will always want lower noseweight values. Generally, keeping the noseweight value around 50% is a good starting point, with 1-2% above and below that being reasonable.

Crossweight is also extremely important, but the dependency on mechanical grip results in crossweight being an extremely sensitive and powerful adjustment. You'll be able to use lower crossweight values at higher grip tracks like Kansas, Michigan, or Las Vegas, since the track's grip can keep the car going straight with the freed-up chassis. Lower grip tracks, or very bumpy tracks where loads under each wheel can change suddenly (like Atlanta's old surface) will need higher crossweight values to keep the car stable. Short tracks depend on crossweight for forward drive, and thus generally run higher than you'd expect. More crossweight will tighten the car through the corner but make it easier to apply the throttle, so drivers will need to find a tradeoff between the two.

REAR ARB

In modern stock cars, rear anti-roll bars are generally outlawed due to the unique ways they were run in 2011-2012. However, in 2003, they were an integral part of the chassis's setup and can be a powerful tool when used.

Rear ARBs are generally used to add right-rear wheel rate in the corners without the need for a high-rate right-rear spring. This helps by giving the chassis the spring rates it needs to navigate a corner without having the stiff rear springs that keep the rear of the car up in the air, generating drag, when going down the straights.

For example, Indianapolis Motor Speedway has very long straights with short, tight corners, so having a rear ARB prevents the need for a large right-rear spring, allowing for less rear spring split and thus better traction off the corners.

Preloading the rear ARB is essential, as the motion ratios for the rear bar are much larger than the front bar. Adding negative preload to the rear bar will keep it loaded as the car enters and leaves the corners, helping the car rotate despite softer rear springs.

